REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

The Quantum Model of the Atom Practice Questions

Grade 12 Advanced | Module 22 Lesson 4 | The Quantum Model of the Atom

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: Photon transitions: $E = hf = hc/\lambda$. Matter-wave relation $\lambda = h/p$. A standing-wave illustration has $2\pi r = n\lambda$. $h = 6.626 \times 10^{-34}$ J s; $c = 3.00 \times 10^{8}$ m/s.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 12 book, printed pages 75–79.

P01. An electron cloud represents:

A. A spatial probability distribution B. A fixed classical orbit with exact position
C. A literal cloud of negatively charged gas D. Only the atomic nucleus

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P02. Stimulated emission occurs when an incident photon:

A. Always ionises the whole atom B. Triggers an excited system to emit a matching photon
C. Is necessarily absorbed without re-emission D. Destroys energy

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P03. Laser light is typically:

A. Necessarily white and incoherent B. Without frequency
C. Highly coherent and directional D. Composed of electrons rather than photons

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P04. Population inversion means:

A. All particles have left the material B. The nucleus becomes negative
C. The atom has zero energy D. More particles occupy the relevant upper laser level than the lower one

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P05. The quantum model describes an electron's location using:

A. Probabilities rather than an exact classical path B. A precisely known circular path at every instant
C. Only the atom's colour D. An arbitrary position unrelated to the state

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P06. A laser emits photons of wavelength 640 nm. Find the energy per photon. Use $h = 6.626 \times 10^{-34}$ J s and $c = 3.00 \times 10^{8}$ m/s.

A. 3.11×10^{-19} J B. 6.21×10^{-19} J
C. 1.55×10^{-19} J D. -3.11×10^{-19} J

P07. A laser pulse contains 0.004 J. Each photon has energy 3.00×10^{-19} J. Find the number of photons.

A. 2.67×10^{16} photons B. 1.33×10^{16} photons
C. 6.67×10^{15} photons D. -1.33×10^{16} photons

P08. In a standing-wave picture, a circular path contains 3 complete de Broglie wavelengths of 0.200 nm each. Find the path circumference.

A. 1.2 nm B. 0.3 nm
C. 0.6 nm D. -0.6 nm

P09. Find the energy of a photon of frequency 2×10^{14} Hz. Use $h = 6.626 \times 10^{-34}$ J s.

A. 2.65×10^{-19} J B. 6.63×10^{-20} J
C. -1.33×10^{-19} J D. 1.33×10^{-19} J

P10. An atomic transition emits a photon of energy 2.5 eV. Calculate its wavelength in nanometres using hc ≈ 1240 eV nm.

A. 496 nm B. 992 nm
C. 248 nm D. -496 nm

P11. What is an orbital in the quantum model?

P12. Distinguish spontaneous from stimulated emission.

P13. State two typical properties of laser light.

P14. Why is energy supplied to a laser medium?

P15. Name one useful laser application and its physical purpose.

P16. A laser emits photons of wavelength 660 nm. Find the energy per photon. Use $h = 6.626 \times 10^{-34}$ J s and $c = 3.00 \times 10^{8}$ m/s.

P17. A laser emits photons of wavelength 680 nm. Find the energy per photon. Use $h = 6.626 \times 10^{-34}$ J s and $c = 3.00 \times 10^{8}$ m/s.

P18. A laser pulse contains 0.006 J. Each photon has energy 3.00×10^{-19} J. Find the number of photons.

P19. A laser pulse contains 0.008 J. Each photon has energy 3.00×10^{-19} J. Find the number of photons.

P20. In a standing-wave picture, a circular path contains 4 complete de Broglie wavelengths of 0.200 nm each. Find the path circumference.

P21. In a standing-wave picture, a circular path contains 5 complete de Broglie wavelengths of 0.200 nm each. Find the path circumference.

P22. Find the energy of a photon of frequency 3×10^{14} Hz. Use $h = 6.626 \times 10^{-34}$ J s.

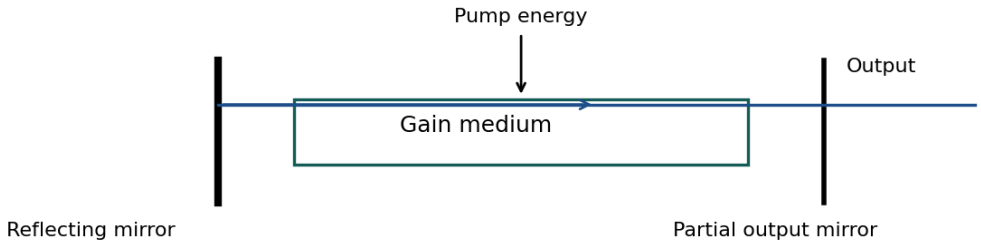
P23. Find the energy of a photon of frequency 4×10^{14} Hz. Use $h = 6.626 \times 10^{-34}$ J s.

P24. An atomic transition emits a photon of energy 3 eV. Calculate its wavelength in nanometres using $hc \approx 1240$ eV nm.

P25. An atomic transition emits a photon of energy 3.5 eV. Calculate its wavelength in nanometres using hc ≈ 1240 eV nm.

P26. Why does the quantum model replace exact Bohr paths with probability distributions?

P27. Explain the role of mirrors in a simple laser cavity.



P28. Compare the emitted photons in stimulated emission with a general spontaneous-emission event.

P29. A learner says a laser creates energy. Correct the statement using an energy balance.

P30. Describe how the standing-wave idea helps explain allowed states qualitatively.